

The turn-division ledger: what happens when the half-turn is replaced by a $1/q$ -turn

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Abstract

Every carry criterion used so far in this campaign reads the phase $\{n/p^j\}$ against $\frac{1}{2}$. That threshold is not intrinsic: it is an artifact of $\binom{2n}{n}$ splitting $2n$ into *two* equal parts. We replace the half-turn by a division of the circle into q equal sectors, prove the corresponding ledger identity, and re-derive every half-turn result at general q . Three consequences. (i) The antipodal law survives verbatim at every q , with threshold $\max(qk, \sqrt{qn})$. (ii) The q -analogue of Erdős #396 is *settled outright* whenever $q \geq k + 1$, by an explicit construction; so the difficulty of #396 is the difficulty of its extreme case $q = 2$, not of the divisibility itself. (iii) Exactly one rigidity survives scrutiny: the registration offset has mean $\frac{1}{2p}$ at *every* division, and the half-turn is distinguished solely by its offset having zero *variance*, which happens iff $\varphi(q) = 1$. Two further candidate rigidities are disposed of — the ceiling maximum is monotonicity in q , and the $\pi/3$ cell is an artifact of the chart.

1 The ledger at a $1/q$ -turn

Throughout $q \geq 2$ is an integer, p a prime, s_p the base- p digit sum, $\{x\}$ the fractional part. Write

$$M_q(n) := \frac{(qn)!}{(n!)^q},$$

so $M_2(n) = \binom{2n}{n}$. $M_q(n)$ is the multinomial coefficient counting the ordered partitions of a qn -set into q blocks of size n , hence a positive integer.

Theorem 1 (turn-division ledger). *For every prime p and every $n \geq 0$,*

$$v_p(M_q(n)) = \sum_{j \geq 1} \left\lfloor q \left\{ \frac{n}{p^j} \right\} \right\rfloor.$$

Proof. By Legendre, $v_p(M_q(n)) = \sum_{j \geq 1} (\lfloor qn/p^j \rfloor - q \lfloor n/p^j \rfloor)$. Writing $x = n/p^j = \lfloor x \rfloor + \{x\}$ gives $\lfloor qx \rfloor = q \lfloor x \rfloor + \lfloor q\{x\} \rfloor$, and the claim follows termwise. $\square$

Verified: 5040 triples (q, p, n) with $2 \leq q \leq 8$ — no mismatch.

At $q = 2$ each summand is the half-turn indicator $\mathbf{1}[\{n/p^j\} \geq \frac{1}{2}]$, taking only the values 0, 1. For $q \geq 3$ the summand is the *index of the sector*: the circle is cut into q equal arcs and the ledger records which arc the phase occupies, a value in $\{0, 1, \dots, q-1\}$. This is the only change, and every statement below is obtained by tracking it.

Corollary 2 (the q -carrier). $p \nmid M_q(n)$ if and only if every base- p digit of n is $< p/q$. Write $G_p^{(q)}$ for this set; $G_p^{(2)}$ is the carrier of #377.

Proof. The j -th summand vanishes iff $\{n/p^j\} < 1/q$, i.e. iff $n \bmod p^j < p^j/q$. Taking $j = 1$ gives the condition on the last digit; applying it to $\lfloor n/p^{j-1} \rfloor$ for each j gives it on every digit, and conversely. $\square$

Verified: 28 000 triples (q, p, n) , $2 \leq q \leq 8$, $p < 30$, $n < 400$ — no mismatch.

The number of admissible digits is $\lceil p/q \rceil$, so the codimension (box-counting cost) of confining a rail to $G_p^{(q)}$ is

$$\beta_p^{(q)} := \frac{\log(p/\lceil p/q \rceil)}{\log p}.$$

A rail is *degenerate* when $\lceil p/q \rceil = 1$, i.e. when $p \leq q$: then $G_p^{(q)} = \{0\}$ and $\beta_p^{(q)} = 1$. Degenerate rails must be excluded from any budget argument; including them returns the spurious optimum $\{2\}$ at every $q \geq 3$.

2 The antipodal law at general q

Theorem 3 (general antipodal law). Let $p > \max(qk, \sqrt{qn})$ be prime and put $\rho = n \bmod p$. Then $v_p(M_q(n)) = \lfloor q\rho/p \rfloor$, and

- (a) if $p \mid n - i$ for some $0 \leq i \leq k$ then $v_p(M_q(n)) = 0$, so p cannot divide $M_q(n)$: extinction;
- (b) if $p \mid n + i$ for some $1 \leq i \leq k$ then $v_p(n + i) = 1 \leq v_p(M_q(n))$: automatic.

Proof. For $j \geq 2$ we have $p^j \geq p^2 > qn \geq n$, so $\{n/p^j\} = n/p^j$ and $q\{n/p^j\} = qn/p^j < 1$; that level contributes 0. Only $j = 1$ survives in Theorem 1, and there $\{n/p\} = \rho/p$, giving $v_p(M_q(n)) = \lfloor q\rho/p \rfloor$. (a) $p \mid n - i$ with $0 \leq i \leq k$ forces $\rho = i$, and $i \leq k < p/q$ because $p > qk$; so $q\rho/p < 1$ and the entry is 0. (b) $p \mid n + i$ with $1 \leq i \leq k$ forces $\rho = p - i$, and $qi \leq qk < p$, so $q\rho/p = q - qi/p > q - 1 \geq 1$ and the entry is at least 1; finally $p^2 > qn \geq n + i$ gives $v_p(n + i) = 1$. $\square$

Verified: for $q = 2, 3, 4, 5, 6$ and $k \leq 6$, $n < 900$: 66 896 negative-shift incidences, all with $v_p(M_q) = 0$, and 50 165 positive-shift incidences, all with $v_p(n + i) = 1 \leq v_p(M_q)$ — no failure in either direction at any q .

Two sharpenings, both of which are invisible at $q = 2$.

Proposition 4 (the positive shift saturates). In case (b) the value is exact: $v_p(M_q(n)) = q - 1$, the maximal sector. Hence $\prod_{1 \leq i \leq k} (n + i)^{q-1}$ divides $M_q(n)$ as far as the primes $p > \max(qk, \sqrt{q(n+k)})$ are concerned.

Proof. $\rho = p - i$ with $1 \leq i \leq k$ and $qk < p$ give $q - 1 < q - qi/p = q\rho/p < q$, so $\lfloor q\rho/p \rfloor = q - 1$. $\square$

Verified: 44 688 incidences over $2 \leq q \leq 7$, $k \leq 6$, $n < 700$ — no failure. So coarsening the division strengthens the positive shift (to the $(q - 1)$ -th power) while leaving the negative shift exactly as dead; at $q = 2$ the exponent is 1 and the asymmetry cannot be seen.

Theorem 5 (*a*-fold extinction). *Let $a \geq 1$, let $p > qk$ be prime with $p^{a+1} > qn$, and suppose $p^a \mid n - i$ for some $0 \leq i \leq k$. Then $v_p(M_q(n)) = 0$.*

Proof. $n \equiv i \pmod{p^a}$ with $i \leq k < p$ gives $n \bmod p^j = i$ for every $j \leq a$, so $\lfloor q\{n/p^j\} \rfloor = \lfloor qi/p^j \rfloor \leq \lfloor qk/p \rfloor = 0$ at each such level; and for $j > a$ we have $p^j \geq p^{a+1} > qn$, so those levels vanish too. $\square$

Verified: 79 479 incidences over $2 \leq q \leq 7$, $k \leq 6$, $n < 900$ — no failure.

Theorem 5 is a *tower* of conditions, not one. Taking $a = 1$ recovers Theorem 3(a) and forces the window to be $(qn)^{1/2}$ -smooth; $a = 2$ forces its square-full part to be $(qn)^{1/3}$ -smooth; $a = 3$ its cube-full part to be $(qn)^{1/4}$ -smooth; and so on. At $q = 2$ with multiplicity one this is invisible, which is why the #396 note recorded only the first floor.

The tower is what kills the obvious constructions, and it does so uniformly in q . The two parametric families that *do* force a smooth window — $n = x^2$ at $k = 1$, with window $\{x^2, (x-1)(x+1)\}$, and the Pell family $y^2 - 2x^2 = -1$, $n = 2x^2 = y^2 + 1$ at $k = 2$ — are built out of squares, and squares are exactly what produces $v_p \geq 2$ at $p \approx \sqrt{n}$, where $p^3 > qn$. Concretely $n = 1682 = 2 \cdot 29^2$ has $v_{29}(n) = 2$ but $v_{29}(M_q(n)) = 0$ for every q from 2 to 14; all eleven Pell terms up to $n \approx 10^{17}$ die this way at every q tested. The $n = x^2$ family survives only when x is itself $(qx^2)^{1/3}$ -smooth. And the Chinese-remainder construction has no reach past $k = 1$ at any q : forcing the window smooth needs $k + 1$ pairwise-coprime moduli of size about $\sqrt{n}$, whose product $n^{(k+1)/2}$ must not exceed n .

So the phase picture is q -independent in form: the divisor condition for a *negative* shift pins ρ to the bottom arc $[0, k]$, which is disjoint from the carry arc $[p/q, p)$ as soon as $k < p/q$; a *positive* shift pins ρ to the top arc $[p-k, p)$, which lies inside the carry arc. Only the threshold moves, from $\max(2k, \sqrt{2n})$ to $\max(qk, \sqrt{qn})$.

3 The non-divisor band at general q

Two earlier notes isolate, at $q = 2$, the primes that fail to divide $\binom{2n}{n}$: the band $(2n/3, n]$ and, more generally, the interval criterion $n \in G_p \iff p(2m+1) \geq 2n+1$ and $p \geq 2m+1$, with $m = \lfloor n/p \rfloor$. Both are the $q = 2$ case of one statement.

Theorem 6 (non-divisor band). *Let $p > \sqrt{qn}$ be prime and put $m = \lfloor n/p \rfloor \geq 1$. Then*

$$p \nmid M_q(n) \iff p(qm+1) > qn \iff p \in I_m^{(q)}(n) := \left(\frac{qn}{qm+1}, \frac{n}{m} \right].$$

The band has width $\frac{n}{m(qm+1)}$, and the top band $m = 1$ is $(\frac{q}{q+1}n, n]$.

Proof. As in Theorem 3, $p^2 > qn$ leaves only the level $j = 1$, so $v_p(M_q(n)) = \lfloor q\{n/p\} \rfloor$. This vanishes iff $\{n/p\} < 1/q$, i.e. iff $n/p - m < 1/q$, i.e. iff $p > n/(m + \frac{1}{q}) = qn/(qm+1)$. The upper endpoint is $m = \lfloor n/p \rfloor$, i.e. $p \leq n/m$. Subtracting the endpoints gives the width. $\square$

Verified: 1 020 921 triples (q, n, p) with $2 \leq q \leq 7$, $n < 1500$, $\sqrt{qn} < p \leq n$ — no mismatch.

At $q = 2$ the condition $p(2m+1) > 2n$ is, for integers, exactly $p(2m+1) \geq 2n+1$, and the side condition $p > \sqrt{2n}$ is the interval criterion's $p \geq 2m+1$; the top band is $(2n/3, n]$.

At $q = 3$ the top band is $(3n/4, n]$, at $q = 4$ it is $(4n/5, n]$: the coarser the division, the narrower the non-divisor bands — consistent with Proposition 14, since a coarser division makes carries easier and non-divisors rarer.

Corollary 7 (criterion at general q). *For $n \geq k + 1$,*

$$\prod_{0 \leq i \leq k} (n-i) \mid M_q(n) \iff (q+1)s_p(n) \geq (k+1) + s_p(n-k-1) + s_p(qn) \text{ for every prime } p.$$

Proof. By Legendre the left side of the valuation inequality is $((k+1) - s_p(n) + s_p(n-k-1))/(p-1)$ and the right side is $(q s_p(n) - s_p(qn))/(p-1)$; clear denominators and collect. $\square$

At $q = 2$ this is the criterion $3s_p(n) \geq (k+1) + s_p(n-k-1) + s_p(2n)$ of the #396 note. *Verified:* reproduces the least solutions 1, 2, 2480, 8178, 45153 for $k = 0, \dots, 4$ at $q = 2$.

4 The q -analogue of #396 is settled for $q \geq k + 1$

Theorem 8. *Let $k \geq 0$ and $q \geq k + 1$. Then $n = k + 1$ satisfies $\prod_{0 \leq i \leq k} (n-i) \mid M_q(n)$.*

Proof. Put $m = k + 1$, so $n = m$ and $\prod_{0 \leq i \leq k} (n-i) = m(m-1) \cdots 1 = m!$. The number of ways to partition a set of qm elements into q unordered blocks of size m is $\frac{(qm)!}{(m!)^q q!}$, an integer; hence $q!$ divides $M_q(m) = \frac{(qm)!}{(m!)^q}$. Since $m \leq q$ we have $m! \mid q!$, so $m!$ divides $M_q(m)$. $\square$

Verified: 104 pairs (q, k) with $2 \leq q \leq 14$, $0 \leq k < q$, by exact integer arithmetic — no failure.

The hypothesis $q \geq k + 1$ is not the exact condition, and the exact condition explains the apparent irregularity at the boundary.

Proposition 9 (when the construction $n = k + 1$ works). *Put $m = k + 1$. Then $\prod_{0 \leq i \leq k} (m-i) \mid M_q(m)$ if and only if*

$$(q+1)s_\ell(m) \geq m + s_\ell(qm) \quad \text{for every prime } \ell.$$

If moreover m is a power of the prime ℓ , the condition at that ℓ reads $q - s_\ell(q) \geq k$; in particular it fails whenever $q \leq k$.

Proof. Substitute $n = m = k + 1$ into Corollary 7, where $n - k - 1 = 0$ and $s_\ell(0) = 0$. If $m = \ell^j$ then $s_\ell(m) = 1$ and $s_\ell(qm) = s_\ell(q\ell^j) = s_\ell(q)$, so the inequality becomes $q + 1 \geq m + s_\ell(q)$, i.e. $q - s_\ell(q) \geq m - 1 = k$. Since $s_\ell(q) \geq 1$ for $q \geq 1$ this forces $q \geq k + 1$. $\square$

Corollary 10 (the boundary $k = q$). *At $k = q$ the construction $n = k + 1$ fails whenever $q + 1$ is a prime power.*

Proof. Take $m = q + 1 = \ell^j$ in Proposition 9: the condition at ℓ is $q - s_\ell(q) \geq k = q$, i.e. $s_\ell(q) \leq 0$, which is false. $\square$

Corollary 10 has a phrasing free of q and k altogether. Putting $n = q + 1$, the criterion at $k = q$ reads $n(s_\ell(n) - 1) \geq s_\ell(n(n - 1))$ for every prime ℓ , and the divisibility being tested is

$$(n!)^n \mid (n(n - 1))! \quad \text{iff} \quad n \text{ is not a prime power.}$$

Verified by exact factorial arithmetic for $2 \leq n \leq 25$ — no mismatch. This is a clean standalone statement and I have *not* run a literature check on it; it should be looked up before anyone calls it new. (The neighbouring fact $(n!)^{n+1} \mid (n^2)!$ is classical, being the count of partitions of an n^2 -set into n unordered n -blocks.)

Measured: the converse also holds throughout $2 \leq q \leq 259$ — at $k = q$ the construction succeeds for exactly those q with $q+1$ *not* a prime power, i.e. $q+1 \in \{6, 10, 12, 14, 15, 18, 20, 21, 22, 24, 26, \dots\}$, and fails for exactly those with $q+1$ a prime power, $q+1 \in \{3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, \dots\}$; 258 values of q , no mismatch, and an independent check to $q = 39$ agrees. The converse is not proved here: the forward direction is Corollary 10, but showing that $s_\ell(m) \geq 2$ for all ℓ suffices needs more than the crude digit bound, since $(\ell - 1)(2 \log_\ell m + 1)$ can exceed m — and the standard submultiplicative bound $s_\ell(ab) \leq s_\ell(a)s_\ell(b)$ is exactly tight at $\ell = m/2$, so it does not close the gap either. The measured margin $\min_\ell [m(s_\ell(m) - 1) - s_\ell(m(m - 1))]$ over all non-prime-power $m < 400$ is always ≥ 0 , with minimum $+2$ attained at $m = 6$.

The law is specific to the boundary $k = q$: at $k = q + 1$ it fails, with 26 mismatches in 118 values, the first at $q = 4$. It is not a general prime-power principle.

Corollary 11. *For every k there is a division q and an n with $\prod_{0 \leq i \leq k} (n - i) \mid M_q(n)$; one may take $q = k + 1$ and $n = k + 1$.*

Table of the least n with $\prod_{0 \leq i \leq k} (n - i) \mid M_q(n)$, computed from Corollary 7 (entries $> 4 \times 10^5$ were not found in the search range):

k	$q=2$	$q=3$	$q=4$	$q=5$	$q=6$	$q=8$	$q=12$
0	1	1	1	1	1	1	1
1	2	2	2	2	2	2	2
2	2480	3	3	3	3	3	3
3	8178	496	4	4	4	4	4
4	45153	13068	496	5	5	5	5
5	.	44255	2627	6	6	6	6
6	.	48512	2628	2628	10	7	7
7	.	48513	48513	48513	10	8	8
8	.	.	.	.	10	10	9
9	.	.	.	.	10	10	10
10	.	.	.	.	.	15	11

The diagonal band of entries $n = k + 1$ is exactly Theorem 8, and it terminates at $k = q - 1$ in every column. Read across a row: the difficulty of #396 is monotone in the *coarseness* of the division, and the half-turn is the worst case because M_2 is the smallest of the M_q and therefore has the least carry budget to spend. Monotonicity in q is provable, not just observed: each sector index $\lfloor q\{n/p^j\} \rfloor$ is nondecreasing in q , so the solution set is upward-closed in q and $n_{\min}(q, k)$ is nonincreasing in q .

The end of the diagonal is a cliff, not a wall. It would be wrong to read the table as saying solutions exist only for $k < q$: they exist well beyond, they are merely large. An independent

search extends the rows to $n_{\min}(2, \cdot) = \dots, 3648841, 7979090, 101130029, 339949252$ for $k = 5, \dots, 8$; $n_{\min}(3, 11) = 321903043$; $n_{\min}(6, 14) = 216715180$. What happens at the end of the diagonal is a jump: at $q = 6$ the value sits at 10 for $k = 6, 7, 8$ and then leaps to 376251 at $k = 9$, a factor of 3.8×10^4 in one step, exactly where $n \leq qk$ fails. Fitted growth of $n_{\min}$ over the last six k in each row: 9.05^k at $q = 2$, 5.50^k at $q = 3$, 4.79^k at $q = 5$, 3.55^k at $q = 6$. So coarsening the division changes the base of the exponential; it does not remove it. Theorem 8 bounds *the reach of one construction*, not the existence of solutions.

5 In what sense the half-turn is special: three candidates, one survivor

Theorem 12 (registration offset: DC and AC). *Write $\lceil p/q \rceil/p = \frac{1}{q} + \frac{r}{qp}$ with $r = (-p) \bmod q$. As p ranges over the primes exceeding q , the value of r runs over $R_q := \{(-a) \bmod q : \gcd(a, q) = 1\}$, a set of size $\varphi(q)$, equidistributed by Dirichlet. Then:*

(DC) *the mean of r over R_q is $q/2$ for every $q \geq 2$, so the mean registration offset is $\frac{1}{2p}$ at every division;*

(AC) *the variance of r over R_q vanishes if and only if $\varphi(q) = 1$, that is $q \in \{1, 2\}$.*

Hence the half-turn is the unique division whose registration offset has zero AC content.

Proof. $q\lceil p/q \rceil = p + r$ gives the displayed identity, and $\gcd(p, q) = 1$ for $p > q$ prime, so by Dirichlet r takes exactly the values R_q . For (DC): $a \mapsto q - a$ maps $(\mathbb{Z}/q)^\times$ to itself and has no fixed point when $q \geq 3$ (a fixed point needs $2a = q$, whence $\gcd(a, q) = a = q/2 > 1$), so R_q is a disjoint union of pairs $\{r, q - r\}$ and its mean is $q/2$; for $q = 2$, $R_2 = \{1\}$ and the mean is $1 = q/2$. For (AC): a multiset with mean $q/2$ has zero variance iff it is the singleton $\{q/2\}$, iff $|R_q| = \varphi(q) = 1$. $\square$

Verified: exact $(\mathbb{E}[r], \text{Var}[r])$ over R_q for $2 \leq q \leq 12$ — $q = 2 : (1, 0)$, $q = 3 : (\frac{3}{2}, \frac{1}{4})$, $q = 4 : (2, 1)$, $q = 5 : (\frac{5}{2}, \frac{5}{4})$, $q = 6 : (3, 4)$, $q = 8 : (4, 5)$, $q = 12 : (6, 13)$ — mean equal to $q/2$ in every case and variance zero only at $q = 2$.

This correction matters, and it reverses an earlier reading of mine. It is *not* true that the offset $\frac{1}{2p}$ is the half-turn's signature: by (DC) that offset is the common mode of every division. What the half-turn alone has is a *fluctuation-free* offset, so that the offsets can be summed rail by rail with no error term; for $q \geq 3$ the fluctuation is carried by the odd Dirichlet characters mod q and no single ledger entry serves all rails.

There is an intrinsic, chart-free companion to (AC). At $q = 2$ the nonzero admissible digits are $\{1, \dots, \frac{p-1}{2}\}$, which is exactly a fundamental domain for $x \mapsto -x$ on $(\mathbb{Z}/p)^\times$: each pair $\{x, -x\}$ meets it once. So the admissible set is not merely an interval, it is a transversal of $(\mathbb{Z}/p)^\times / \{\pm 1\}$, for every odd p . Both (AC) and this statement are avatars of the same involution $x \mapsto -x$: $\varphi(q) = 1$ says $-1 \equiv 1 \pmod{q}$, and $\{\pm 1\}$ is the subgroup whose cosets an interval can cut out. For $q \geq 3$ no such group-theoretic description of the admissible interval exists except sporadically.

Theorem 13 (the limiting cost angle). Define $\theta_p^{(q)} \in [0, \pi/2)$ by $\cos \theta_p^{(q)} = \lceil p/q \rceil / p$, so that $\beta_p^{(q)} = \log \sec \theta_p^{(q)} / \log p$, and let $\theta_\infty^{(q)} = \lim_{p \rightarrow \infty} \theta_p^{(q)} = \arccos(1/q)$. Then $\theta_\infty^{(q)}$ is a rational multiple of π if and only if $q \in \{1, 2\}$, and for $q = 2$ it equals $\pi/3$.

Proof. $\cos \theta_\infty^{(q)} = 1/q$ is rational, so by Niven's theorem — the only rational values of $\cos \theta$ at θ a rational multiple of π are $0, \pm \frac{1}{2}, \pm 1$ — we need $1/q \in \{0, \frac{1}{2}, 1\}$, i.e. $q \in \{1, 2\}$. At $q = 2$, $\arccos \frac{1}{2} = \pi/3$. $\square$

The theorem is true, and its content is nil. The deduction uses only “ $1/q = \frac{1}{2} \iff q = 2$ ”: any rational-valued family meets the five-element Niven set exactly where it equals one of five numbers, and Niven merely re-expresses that as a statement about angles. The apparent payoff — that the half-turn lands on the μ_6 cell — is an artifact of the chart, for four reasons.

(i) *Other equally natural charts give other cells, or none.* Taking $\sin \theta = 1/q$ also selects $q = 2$ but at $\theta = \pi/6$, the μ_{12} cell, not μ_6 . Taking $\tan \theta = 1/q$ selects nothing, since Niven's tangent set is $\{0, \pm 1\}$ and forces $q = 1$. The density $1/q$ read bare supports no lattice statement at all. And the codimension $\beta_p^{(q)}$ is irrational for every nondegenerate pair (p, q) : if $\log(p/a)/\log p = u/v$ then $a^v = p^{v-u}$, so a is a power of p , and $1 \leq a \leq p$ forces $a = 1$ ($\beta = 1$) or $a = p$ ($\beta = 0$). So β never lands on any lattice, at any q . Perturbing the family decouples the two conditions further: $\cos \theta = 1/(q-1)$ selects $\{2, 3\}$ with $q = 3$ on μ_6 , and $\cos \theta = (q-1)/(q+1)$ selects $\{3\}$ alone.

(ii) *The object supplies its own angle, and it distinguishes nothing.* The admissible digits are precisely the p -th roots of unity lying in the arc $[0, 2\pi/q)$. That arc has width $2\pi/q$, a rational multiple of π for every q ; in the construction's own angular coordinate each division sits on μ_q and no division is special.

(iii) *The exact cell is occupied only by the trivial rail.* Since $\delta = \lceil p/q \rceil / p \in (0, 1]$, the values $0, -\frac{1}{2}, -1$ are unreachable; $\delta = 1$ iff $q = 1$; and $\delta = \frac{1}{2}$ needs $2\lceil p/q \rceil = p$, hence p even, hence $p = 2$. So the complete list of exact Niven landings is: $q = 1$ with every p , and $p = 2$ with every $q \geq 2$, where $\delta = \frac{1}{2}$ and $\theta = \pi/3$ exactly. The sole exact μ_6 landing in the whole family is therefore the degenerate rail $p = 2$, and it occurs at every division.

(iv) *The 6 has no preimage in the object.* At $q = 2$ the construction produces 2 everywhere — arc width π , stabiliser $\{\pm 1\}$, offset modulus 2. The 6 enters only as the order of the root of unity attached to $\arccos \frac{1}{2}$, i.e. from the chart. A cell whose order cannot be exhibited anywhere in the object is not structure.

What survives is the bare selection: $\arccos(1/q)/\pi \notin \mathbb{Q}$ for every $q \geq 3$. The cell does not.

Proposition 14 (the ceiling collapses). Let $\Lambda(q) = \max\{\sum_{p \in S} 1/p : S \text{ a set of nondegenerate rails with } \sum_{p \in S} 1\}$. Over rails $p < 300$ and $|S| \leq 5$ the computed values are

q	2	3	4	5	6	7
S	$\{3, 5, 7\}$	$\{5, 11\}$	$\{5, 23\}$	$\{7, 89\}$	$\{7, 139\}$	$\{11, 827\}$
Λ	0.676190	0.290909	0.243478	0.154093	0.150051	0.092118
q	8	9	10	11	12	
S	$\{11, 1321\}$	$\{11, 1993\}$	$\{11, 2851\}$	$\{13, 7109\}$	$\{13, 9817\}$	
Λ	0.091666	0.091411	0.091260	0.077064	0.077025	

Every entry is confirmed by an independent branch-and-bound search in ratio-sorted order out to $p \leq 5000$. The supremum is not attained (see below), so “max” should be read as “sup”.

Three cautions, two of which retract earlier readings.

A search cap of $p < 300$ is far too small. An earlier version of this table reported the single rail $\{11\}$ for $7 \leq q \leq 10$ and $\{13\}$ for $q = 11, 12$; every one of those rows was wrong, the second rail merely sitting between 827 and 9817. The values above come from branch-and-bound over $p < 20\,000$ with a rigorous efficiency bound, cross-checked against unpruned enumeration. Likewise *it is false that only $q = 2$ supports three rails*: at $q = 3$ the four rails $\{191, 193, 197, 199\}$ are feasible within $p < 200$ at cost 0.8276. What is true is only that $q \geq 3$ cannot reach three rails *cheaply*.

Coarsening the division destroys monotonicity of rail cost in p , and this is the AC content of Theorem 12 made visible. At $q = 2$ the cost $\beta_p^{(2)}$ is strictly decreasing in p — there are no inversions among primes below 200 000, and none can occur, since $\frac{d}{dp}\beta$ has the sign of $\frac{\log p}{p+1} - \log \frac{2p}{p+1}$, which is negative for $p \geq 3$ because the first term is at most $\frac{\log 3}{4} = 0.275$ and the second at least $\log \frac{3}{2} = 0.406$. For $q \geq 3$ inversions are everywhere: 3271 of them at $q = 3$ below 200 000, 4128 at $q = 4$, 7142 at $q = 12$. The consequence is concrete. At $q = 4$,

$$\beta_{17} = 0.431939, \quad \beta_{19} = 0.453397, \quad \beta_{23} = 0.428556,$$

so $\{5, 17\}$ costs 1.001262 and $\{5, 19\}$ costs 1.022721 — both infeasible — while $\{5, 23\}$ costs 0.997879 and is feasible. The optimal second rail is the *largest* of the three, chosen not by size but by $23 \equiv -1 \pmod{4}$ giving the small offset $r = 1$. That is exactly what a fluctuating registration offset does: it reorders the rails without moving the ceiling much.

The maximum at $q = 2$ is monotonicity, not rigidity. $\lceil p/q \rceil$ is nonincreasing in q , so $\beta_p^{(q)}$ is nondecreasing in q for every fixed p (verified, 3596 pairs, no violation); the feasible region therefore shrinks monotonically and Λ can only decrease. Since $q = 2$ is the smallest admissible division it is automatically the maximiser, as it would be for any monotone family. This is why Proposition 14 is not offered as a rigidity.

No nontrivial optimum is known to be attained. Every nondegenerate $\beta_p^{(q)}$ is irrational, by the argument in (i) of the discussion following Theorem 13. That alone does *not* settle the question — a sum of irrationals can be rational — and whether $\sum_{p \in S} \beta_p^{(q)} = 1$ is possible for a nondegenerate set S is left open here: the terms are $\log(p/a)/\log p$ with *different* denominators, so this is not a rational linear form in logarithms and Baker’s theorem does not apply directly. Numerically the sum never came within 10^{-11} of 1 over the searched range. Granting that, each optimum is a supremum approached through ever longer far tails. At $q = 2$ the tail is exact: the last admissible rail is $P = 398\,328\,266\,023$, contributing $1/P = 2.5104921877 \times 10^{-12}$ and giving $\Lambda(2) = 0.6761904761929866826639\dots$; a further rail would need $P' \approx 1.75 \times 10^{8602545911616}$. Coarser divisions have tails too ($q = 3$ needs $P \approx 1.5 \times 10^{54}$, $q = 4$ needs $P \approx 8 \times 10^{283}$), so the far tail is generic and not a feature of the half-turn. By contrast the spurious optimum $\{2\}$ at $q \geq 3$ had $\sum \beta = 1$ *exactly* and *was* attained — the cleanest signature that a degenerate rail has been admitted.

The one residue here that is not monotone: $q = 2$ is the only division with $\Lambda(q) > 1/q$, by a wide margin ($0.676 = 1.35 \times \frac{1}{2}$, against $\Lambda(q) < 1/q$ for every $3 \leq q \leq 10$). This traces to $p = 3$ being the half-turn’s smallest nondegenerate rail, where the offset correction is large: $\beta_3^{(2)} = 0.369$ against a leading term $\log 2/\log 3 = 0.631$.

6 The window lemma at a $1/q$ -turn: the arc fan

The window lemma behind #683, #685 and #1094 is stated for $p > n/2$ with $k \leq n/2$. Both halves are the half-turn, and at a $1/q$ -turn the single window opens into a fan.

Theorem 15 (arc fan). *Let $q \geq 2$, $n \geq q^2$, $1 \leq k \leq n/q$, and let $p > n/q$ be prime. Put $j = \lfloor n/p \rfloor$, so $0 \leq j \leq q-1$. Then*

$$v_p \binom{n}{k} = \begin{cases} 1, & j \geq 1 \text{ and } jp > n - k, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently $\{p > n/q : p \mid \binom{n}{k}\}$ is the disjoint union over $j = 1, \dots, q-1$ of the primes in $(\frac{n-k}{j}, \frac{n}{j}]$. Arc j has length exactly k/j and lies inside the band $(\frac{n}{j+1}, \frac{n}{j}]$.

Proof. $n \geq q^2$ gives $p > n/q \geq \sqrt{n}$, so $p^2 > n$ and only level 1 survives; $k \leq n/q < p$ gives $\lfloor k/p \rfloor = 0$, so $v_p \binom{n}{k} = \lfloor n/p \rfloor - \lfloor (n-k)/p \rfloor$, the number of multiples of p in $(n-k, n]$. That interval has length $k < p$, so the count is 0 or 1, and is 1 exactly when the largest multiple $jp \leq n$ exceeds $n-k$. Finally $p > n/q$ forces $j \leq q-1$, and $k \leq n/q \leq n/(j+1)$ places arc j inside its band. $\square$

At $q = 2$ only $j = 1$ survives and Theorem 15 is the classical window lemma. So the single window becomes a fan of $q-1$ arcs, the j -th of length k/j at height n/j . The refined window is already turn-free: for $p > \max(k, \sqrt{n})$ one has $p \mid \binom{n}{k} \iff (n \bmod p) < k$, with $v_p = 1$; it is the fan, not the refinement, that carries the q -dependence.

Theorem 16 (the ledger is Boolean only at the half-turn). *For every prime $p > n/q$,*

$$v_p \binom{n}{k} \leq \lceil \log_2 q \rceil,$$

and the bound is attained. In particular $v_p \binom{n}{k} \in \{0, 1\}$ for all $p > n/q$ if and only if $q \leq 2$.

Proof. $v_p \geq m$ requires $p^m \leq n$, while $p > n/q$ gives $n < qp$; hence $p^m < qp$, so $p^{m-1} < q$, so $2^{m-1} < q$ and $m \leq \lceil \log_2 q \rceil$. Sharpness: take $p = 2$, $k = 1$, $n = 2^{\lceil \log_2 q \rceil}$. $\square$

Verified: the maximum of $v_p \binom{n}{k}$ over $p > n/q$, all $n < 220$ and $k \leq n/2$, is 1, 2, 2, 3, 3, 3, 3, 4, 4 for $q = 2, \dots, 10$ — equal to $\lceil \log_2 q \rceil$ in every case, with no violation.

This is the one statement in this note that distinguishes $q = 2$ *intrinsically and qualitatively*: at the half-turn the binomial ledger above the threshold is a single bit, and at every coarser turn it is a staircase. Unlike Proposition 14 it is not an extremum of a monotone quantity, and unlike Theorem 13 it involves no chart. It is still a threshold in a monotone family — $\lceil \log_2 q \rceil$ is nondecreasing — so the phrase “unique largest” is doing the work; what is not automatic is that the value 1 is attained at all. Correspondingly, $v_p \geq 2$ with $p > n/q$ forces $n < q^2$: at a $1/q$ -turn higher valuations live only in the low range.

The multinomial side is the same lemma one level up. For $M = A!/(a_1! \cdots a_q!)$ with $A = \sum_i a_i$, the level- j entry $\lfloor (\sum_i (a_i \bmod p^j))/p^j \rfloor$ is the *wrap number* of the q residues on the circle $\mathbb{Z}/p^j$, valued in $\{0, \dots, q-1\}$; and for $p > \max(\max_i a_i, \sqrt{A})$ one gets $v_p(M) = \lfloor A/p \rfloor$, taking the value m exactly on $p \in (\frac{A}{m+1}, \frac{A}{m}]$. At $q = 2$ with $a = (k, n-k)$ this is $v_p = \lfloor n/p \rfloor \in \{0, 1\}$, the classical window lemma again: the window *value* 1 becomes a staircase running through every value $0, \dots, q-1$.

7 Erdős #376 at general q

#376 asks whether $\binom{2n}{n}$ is coprime to $105 = 3 \cdot 5 \cdot 7$ infinitely often; equivalently, by Corollary 2, whether infinitely many n lie in $G_3^{(2)} \cap G_5^{(2)} \cap G_7^{(2)}$. The q -analogue collapses immediately.

Proposition 17. *For every $q \geq 3$ the only $n \geq 0$ with $M_q(n)$ coprime to 105 is $n = 0$.*

Proof. $3 \leq q$, so $\lceil 3/q \rceil = 1$ and the only admissible base-3 digit is 0; hence $G_3^{(q)} = \{0\}$ and $3 \mid M_q(n)$ for every $n \geq 1$. $\square$

So the three-rail problem is open at $q = 2$ and refuted for every $q \geq 3$, with no intermediate regime: there is no coarser division at which #376 is easier, because the smallest rail dies first. The per-rail dimensions $d_p^{(q)} = 1 - \beta_p^{(q)} = \log \lceil p/q \rceil / \log p$ make this quantitative. At $q = 2$ they are 0.630930, 0.682606, 0.712414 summing to 2.025950; at $q = 3$, 0, 0.430677, 0.564575 summing to 0.995252; at $q = 4$, 0.786884; at $q = 5, 6$, 0.356207; at $q \geq 7$, zero.

Two thresholds are easy to conflate here and should not be. The condition $\sum_p d_p > 1$ is the *two-rail* criterion; three simultaneous digit constraints add codimensions, so the right test is $\sum_p d_p > 2$, equivalently $\sum_p \beta_p < 1$. Only $q = 2$ passes it, with predicted dimension $\sum d_p - 2 = 0.025950$. The $q = 3$ value 0.995252 is *not* a near miss of the three-rail test — it misses by 1.005. Note also the bookkeeping identity $1 - \sum_{p \in \{3,5,7\}} \beta_p^{(2)} = \sum_p d_p - 2$: the leftover budget in Proposition 14 and the predicted #376 dimension are the same number, 0.025950322273.

Verified: $|G_p^{(q)} \cap [0, p^J]| = \lceil p/q \rceil^J$ exactly, for $p \in \{3, 5, 7, 11\}$, $2 \leq q \leq 8$, $1 \leq J \leq 4$ — no failure. So the per-rail box count is exact at every scale and nothing heuristic enters below the multi-rail level; the budget rule “ $\sum \beta < 1 \Rightarrow$ infinitely many” is precisely the assumption that the digit constraints at distinct bases are independent, and that assumption is the whole content of #376.

Remark 18 (attribution of the budget rule). *The rule “ $\sum_p \beta_p^{(q)} < 1 \Rightarrow$ infinitely many” is not ours. With $\kappa_j = 1/q$ it is verbatim Conjecture 1 of Bloom and Croot, Integers with small digits in multiple bases, arXiv:2509.02835: for bases g_j with associated $\kappa_j \in (0, 1]$, if $\sum_j \log_{g_j} (g_j / \lceil \kappa_j g_j \rceil) < 1$ then there are infinitely many n all of whose base- g_j digits are $< \kappa_j g_j$ — and $\lceil \kappa_j g_j \rceil$ is exactly our $\lceil p/q \rceil$. They compute the Graham case $\log_3 \frac{3}{2} + \log_5 \frac{5}{3} + \log_7 \frac{7}{4} = 0.974\dots$, our 0.974050. The finiteness clause and the predicted dimension $\sum_p d_p - (r - 1)$ are Conjecture 1.4 of Han Yu, Fractal projections with an application in number theory, arXiv:2004.05924, stated there for $\kappa = \frac{1}{2}$. The exponent 0.02595 is Pomerance’s published heuristic (Amer. Math. Monthly **122** (2015), §4), recorded on OEIS A030979. What the q -family contributes is only the dictionary $\kappa = 1/q \leftrightarrow (qn)!/(n!)^q$, connecting the digit side to a multinomial; that is a dictionary entry, not a theorem. Note also where Remark 19 sits: Bloom–Croot have general κ but only the infinitude clause, Han Yu has the finiteness clause but only at $\kappa = \frac{1}{2}$, and the $q = 3$ two-rail case needs both at once — it is the uncovered corner between two printed conjectures.*

Remark 19 (a sharper critical case than #376 itself). *At $q = 3$ the two smallest usable rails give $\beta_5^{(3)} + \beta_7^{(3)} = 1.004748407873$, missing the budget by 0.475% — the thinnest margin anywhere in the family. There is a sharper way to say this. The only unconditional two-base theorem in the literature is Theorem 1 of Erdős, Graham, Ruzsa and Straus, On the prime*

factors of $\binom{2n}{n}$, *Math. Comp.* **29** (1975), 83–92, p. 84: if $\frac{A}{p-1} + \frac{B}{q-1} > 1$ then infinitely many integers have all base- p digits $< A$ and all base- q digits $< B$. Our case is base 5 with digits $< 5/3$, i.e. $A = 2$, and base 7 with digits $< 7/3$, i.e. $B = 3$, giving

$$\frac{A}{p-1} + \frac{B}{q-1} = \frac{2}{4} + \frac{3}{6} = \frac{1}{2} + \frac{1}{2} = 1$$

exactly — failing the strict inequality by zero. So this is not merely a near miss of a heuristic budget; it is the exact boundary case of the one theorem that would settle it. (Bloom and Croot restate EGRS Theorem 1 in the inequivalent form $\frac{[\kappa_1 g_1]-1}{g_1-1} + \frac{[\kappa_2 g_2]-1}{g_2-1} \geq 1$, under which our case gives $\frac{1}{4} + \frac{2}{6} = 0.583$; the case is uncovered either way, but the two forms must not be quoted interchangeably.) The corresponding question, whether $(3n)!/(n!)^3$ is coprime to 35 infinitely often, is therefore predicted finite, and barely. A search to 2.98×10^{17} finds exactly nine solutions,

$$0, 1, 750, 751, 3150, 3151, 4761974610150, 4761974610155, 4761974610156,$$

with a desert of nine orders of magnitude between 3151 and 4.76×10^{12} that then ends. Near-critical verdicts of this kind are not numerically decidable — at $q = 4$ the pair $\{5, 23\}$ has predicted dimension 0.00212 and its solution count went from 9 to 30 between 10^9 and 3×10^{17} , in tight clusters, where the prediction allows a factor 1.04 — so this is a prediction on record, not a result.

Remark 20 (do not rewrite the digit criterion with a floor). Corollary 2 must be kept in its strict form, “every base- p digit $< p/q$ ”. The tempting rewrite “every digit $\leq \lfloor p/q \rfloor$ ” is false exactly when $p = q$: there $\lfloor p/q \rfloor = 1$ but $p/q = 1$, so the strict form correctly forces every digit to 0, matching $G_p^{(p)} = \{0\}$, while the floor form wrongly admits the digit 1. At $p = q = 2$ the floor version would assert that $\binom{2n}{n}$ is always odd. The strict form is valid for all primes with no side condition, and the alphabet size $\lceil p/q \rceil$ is likewise universally correct, so $\beta_p^{(q)}$ has no exceptional regime.

Remark 21 (how much of this is really rigidity). The three statements above all return $q \in \{1, 2\}$, and it would be easy to read that agreement as threefold confirmation. It is not. Proposition 14 is monotonicity in q and would hold for any monotone family; Theorem 13 is true with nil content and its cell is chart-created. Only Theorem 12(AC) is neither an extremum nor a convention.

Are (AC) and Niven the same statement twice? No, and the near miss is instructive. Niven’s theorem is a φ -condition — $2\cos(2\pi k/n)$ generates $\mathbb{Q}(\zeta_n)^+$, of degree $\varphi(n)/2$, so rationality is $\varphi(n) \leq 2$, i.e. $n \in \{1, 2, 3, 4, 6\}$ — but on a different modulus. Niven’s modulus is the order $n = 6$ of the root of unity attached to $\cos \theta = \frac{1}{2}$, with $\varphi(6) = 2$: an index condition, $(\mathbb{Z}/n)^\times = \{\pm 1\}$. Ours is $q = 2$ with $\varphi(2) = 1$: an order condition, $(\mathbb{Z}/q)^\times = \{1\}$. That the two cut out the same set $\{1, 2\}$ is extensional. Perturbing the family separates them: with $\delta = 1/(q-1)$ Niven selects $\{2, 3\}$ and φ selects $\{2\}$; with $\delta = (q-1)/(q+1)$ Niven selects $\{3\}$ and φ selects $\{2\}$ — disjoint.

They do share an ancestor, the involution $x \mapsto -x$: $\varphi(q) = 1$ says $-1 \equiv 1 \pmod{q}$, and $\varphi(n) \leq 2$ says $(\mathbb{Z}/n)^\times = \{\pm 1\}$. But ours acts on the digits, and is therefore intrinsic to the object; Niven’s acts on the cyclotomic field of the chart’s angle, and is not. Only one of the two ancestors is present in the construction.

The claim worth carrying forward is therefore the narrow one: the mean registration offset is $\frac{1}{2p}$ at every division, and the half-turn is the unique division at which that offset does not fluctuate.

Remark 22 (harmonic content). *The object's own scale is the circle cut into q sectors, never unit 1. The DC mode is the sector index $\lfloor q\{n/p^j\} \rfloor$, evaluated exactly in Theorem 1 with no remainder; the entire AC content at a prime above the threshold is the single sector occupied by p/p . The difficulty lives on the carrier, in the smooth window that Theorem 3(a) forces, and Theorem 8 shows that difficulty is purchased by the coarseness of the division: at $q \geq k + 1$ the carry budget of M_q covers the window outright.*

Remark 23 (register). **Proved:** Theorem 1, Corollary 2, Theorem 3, Corollary 7, Theorem 8, Theorem 12, Theorem 13. All are elementary from Legendre, Kummer, Dirichlet and Niven, and every one was checked numerically before being written. **Measured:** Proposition 14 and the least- n table, both within the stated search ranges. **Not proved:** Erdős #396 itself, i.e. the case $q = 2$ for every k ; and, for each fixed $q \geq 3$, the analogue for every k — Theorem 8 covers only $k < q$. **Falsifier:** a prime $p > \max(qk, \sqrt{qn})$ dividing some $n - i$ with $v_p(M_q(n)) > 0$ would refute Theorem 3(a); none was found in 66 896 tested incidences across five values of q . A division $q \geq 3$ with constant registration offset would refute Theorem 12; none exists, by $\varphi(q) \geq 2$.